Online Appendix – Not For Publication

Investment Irreversibility in a Granular World

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A Analytical Results

This appendix presents analytical results from a simple, partial-equilibrium model that illustrates the relationship between cross-sectional capital allocation and aggregate fluctuations in an economy with a finite number of firms. Relative to the body of the paper, the key simplification is that the model abstracts from investment irreversibility—this preserves an analytical solution while leaving room to discuss, qualitatively, how irreversibility might amplify or dampen the mechanisms identified here. The key extension relative to a purely capital-only formulation is that we let firms hire labor at an exogenous wage. The wage shifts the mean of log per-capita output deterministically, without altering the productivity–capital component of cross-sectional variance and its role in granular fluctuations developed in the body.

A.1 Firm Problem with Labor

Each of $M \in (0, \infty)$ firms uses capital and labor to produce a homogeneous good with the decreasing-returns Cobb-Douglas technology

$$y_i = \varepsilon_i k_i^{\alpha} n_i^{\nu}, \qquad \alpha + \nu < 1, \tag{A.1}$$

where k_i is predetermined at the beginning of the period and n_i is chosen statically. Idiosyncratic productivity follows

$$\ln \varepsilon_{i,t+1} = \xi_{i,t} + \epsilon_{i,t+1}, \qquad \xi_{i,t} \sim N(0, \sigma_{\xi}^2), \quad \epsilon_{i,t+1} \sim N(0, \sigma_{\epsilon}^2), \tag{A.2}$$

with $\xi_{i,t}$ observed at time t and $\epsilon_{i,t+1}$ realized at the start of $t+1$, both independent across firms and over time. Taking the wage ω as exogenous, the labor first-order condition yields

$$n_i = \left(\frac{\nu \varepsilon_i k_i^{\alpha}}{\omega} \right)^{\frac{1}{1-\nu}}. \tag{A.3}$$

Substituting (A.3) into (A.1) gives firm output as

$$y_i = (\varepsilon_i k_i^{\alpha})^{\frac{1}{1-\nu}} \left(\frac{\nu}{\omega}\right)^{\frac{\nu}{1-\nu}} . \tag{A.4}$$

It is convenient to define rescaled firm output

$$\tilde{y}_i \equiv y_i \cdot \left(\frac{\nu}{\omega}\right)^{-\frac{\nu}{1-\nu}} = (\varepsilon_i k_i^{\alpha})^{\frac{1}{1-\nu}} , \tag{A.5}$$

which strips out the deterministic wage scaling and isolates the productivity–capital component.

The firm's dynamic problem is to choose an investment plan $\{i_{i,t+j}\}_{j\geq 0}$ to maximize the present discounted value of expected profits net of investment, given a constant interest rate r and depreciation rate δ:

$$\max \ \mathbb{E}_t \sum_{j=0}^{\infty} \frac{1}{(1+r)^j} \left(\pi(\varepsilon_{i,t+j}, k_{i,t+j}; \omega) - i_{i,t+j}\right), \quad \text{s.t.} \quad k_{i,t+j+1} = (1-\delta)k_{i,t+j} + i_{i,t+j}, \tag{A.6}$$

where $\pi(\varepsilon, k; \omega)$ is static profit after labor optimization,

$$\pi(\varepsilon, k; \omega) = (1-\nu)\left(\frac{\nu}{\omega}\right)^{\frac{\nu}{1-\nu}} (\varepsilon k^{\alpha})^{\frac{1}{1-\nu}} . \tag{A.7}$$

A.2 Lemma 1

Lemma 1. Suppose $(\ln \varepsilon_i, \ln k_i)$ is jointly normal with means (μ_ε, μ_k), variances $(\sigma_\varepsilon^2, \sigma_k^2)$, and covariance $\sigma_{\varepsilon k}$. Then $\ln \tilde{y}_i \sim N(\mu_{\tilde{y}}, \sigma_{\tilde{y}}^2)$ with

$$\mu_{\tilde{y}} = \tfrac{1}{1-\nu} \left(\mu_\varepsilon + \alpha \mu_k\right), \tag{A.8}$$

$$\sigma_{\tilde{y}}^2 = \tfrac{1}{(1-\nu)^2} \left(\sigma_\varepsilon^2 + \alpha^2 \sigma_k^2 + 2\alpha\, \sigma_{\varepsilon k}\right). \tag{A.9}$$

Equivalently, $\ln y_i \sim N\big(\mu_{\tilde{y}} + \tfrac{\nu}{1-\nu} \ln(\nu/\omega),\ \sigma_{\tilde{y}}^2\big)$: the wage shifts the mean of $\ln y_i$ deterministically but does not affect its variance.

Proof. See Appendix A.4. ■

The covariance $\sigma_{\varepsilon k}$ enters $\sigma_{\tilde{y}}^2$ linearly with positive coefficient $2\alpha/(1-\nu)^2$. Higher productivity–

capital alignment thus raises the cross-sectional variance of (rescaled) firm output, which is what drives the granularity result that follows.

A.3 Proposition 1

Proposition 1. Aggregate per-capita output is $\bar{Y} = \frac{1}{M}\sum_{i=1}^{M} y_i$. Under Lemma 1, $\bar{Y}$ is approximately log-normal with coefficient of variation

$$\mathrm{CV}(\bar{Y}) = \frac{\sqrt{\mathrm{Var}(\bar{Y})}}{\mathbb{E}(\bar{Y})} = \sqrt{\frac{e^{\sigma_{\tilde{y}}^2} - 1}{M}}, \tag{A.10}$$

where $\sigma_{\tilde{y}}^2$ is given by (A.9). Equivalently,

$$\ln \bar{Y} \;=\; \text{constant} - \tfrac{\nu}{1-\nu}\ln\omega + \zeta, \qquad \zeta \sim N\!\left(0,\, \ln\!\left[\tfrac{e^{\sigma_{\tilde{y}}^2}-1}{M} + 1\right]\right). \tag{A.11}$$

Proof. See Appendix A.5. ■

The result has three implications worth highlighting.

Granularity. For finite M, $\mathrm{CV}(\bar{Y}) > 0$: idiosyncratic shocks do not average out. As $M \to \infty$, $\mathrm{CV}(\bar{Y}) \to 0$, recovering the law of large numbers.

Productivity–capital covariance. $\mathrm{CV}(\bar{Y})$ is increasing in $\sigma_{\varepsilon k}$. Cross-sectional alignment between productivity and capital amplifies aggregate fluctuations; misalignment dampens them. Equivalently, the covariance can be measured through the cross-sectional dispersion of marginal products of capital, which provides an empirical handle on $\sigma_{\varepsilon k}$.

Wage variation. The wage enters $\ln \bar{Y}$ linearly with coefficient $-\nu/(1-\nu)$. Wage fluctuations therefore translate one-for-one into fluctuations in log per-capita output, scaled by $\nu/(1-\nu)$.

A.4 Proof of Lemma 1

The proof has two stages: (i) establishing joint log-normality of $(\ln \varepsilon_{i,t+1}, \ln k_{i,t+1})$ from the firm's optimal capital choice with labor, and (ii) computing the moments of $\ln \tilde{y}_i$ from

the joint distribution.

Stage 1: joint log-normality. The reduced-form profit (A.7) gives

$$\pi_k(\varepsilon, k; \omega) = \alpha(\nu/\omega)^{\nu/(1-\nu)} \varepsilon^{1/(1-\nu)} k^{\alpha/(1-\nu)-1}.$$

Combining the dynamic Bellman associated with (A.6) with the envelope condition gives the capital first-order condition $\mathbb{E}_t[\pi_k(\varepsilon_{i,t+1}, k_{i,t+1}; \omega)] = r + \delta$, i.e.,

$$\alpha \Big(\frac{\nu}{\omega}\Big)^{\frac{\nu}{1-\nu}} \mathbb{E}_t \left[\varepsilon_{i,t+1}^{\frac{1}{1-\nu}} \right] k_{i,t+1}^{\frac{\alpha}{1-\nu}-1} = r + \delta. \tag{A.12}$$

Since $\alpha + \nu < 1$, the exponent $1 - \alpha/(1-\nu) = (1-\nu-\alpha)/(1-\nu)$ is positive, and (A.12) can be solved for $k_{i,t+1}$. Using the productivity process and the log-normal moment formula $\mathbb{E}_t[\varepsilon_{i,t+1}^{1/(1-\nu)}] = \exp(\xi_{i,t}/(1-\nu) + \sigma_\epsilon^2/[2(1-\nu)^2])$, taking logs gives

$$\ln k_{i,t+1} = A + B\, \xi_{i,t}, \qquad B = \frac{1}{1-\nu-\alpha}, \tag{A.13}$$

where A collects deterministic constants in $r, \delta, \alpha, \nu, \omega, \sigma_\epsilon^2$. The slope reduces to $1/(1-\alpha)$ when $\nu \to 0$, recovering the capital-only case. Because $(\xi_{i,t}, \epsilon_{i,t+1})$ is jointly normal and $\ln k_{i,t+1}$ is affine in $\xi_{i,t}$, $(\ln \varepsilon_{i,t+1}, \ln k_{i,t+1})$ is jointly normal with

$$\sigma_k^2 = \frac{\sigma_\xi^2}{(1-\nu-\alpha)^2}, \tag{A.14}$$

$$\sigma_{\varepsilon k} = \frac{\sigma_\xi^2}{1-\nu-\alpha}, \tag{A.15}$$

$$\sigma_\varepsilon^2 = \sigma_\xi^2 + \sigma_\epsilon^2. \tag{A.16}$$

Stage 2: moments of $\ln \tilde{y}_i$. From (A.5),

$$\ln \tilde{y}_i = \tfrac{1}{1-\nu} \left(\ln \varepsilon_i + \alpha \ln k_i \right), \tag{A.17}$$

which is a linear combination of jointly normal variables and is therefore normal. Direct calculation gives the mean and variance stated in the lemma:

$$\mu_{\tilde{y}} = \tfrac{1}{1-\nu}(\mu_\varepsilon + \alpha\mu_k)\,, \tag{A.18}$$

$$\sigma^2_{\tilde{y}} = \tfrac{1}{(1-\nu)^2}\big(\sigma^2_\varepsilon + \alpha^2\sigma^2_k + 2\alpha\,\sigma_{\varepsilon k}\big)\,. \tag{A.19}$$

■

A.5 Proof of Proposition 1

By Lemma 1, $y_i \sim LN(\mu_y, \sigma^2_{\tilde{y}})$ with $\mu_y = \mu_{\tilde{y}} + \frac{\nu}{1-\nu}\ln(\nu/\omega)$, i.i.d. across firms. The first two moments of a log-normal variable are

$$\mathbb{E}(y_i) = e^{\mu_y + \sigma^2_{\tilde{y}}/2}, \qquad \mathrm{Var}(y_i) = (e^{\sigma^2_{\tilde{y}}} - 1)\, e^{2\mu_y + \sigma^2_{\tilde{y}}}. \tag{A.20}$$

Independence and linearity give

$$\mathbb{E}(\bar{Y}) = e^{\mu_y + \sigma^2_{\tilde{y}}/2}, \qquad \mathrm{Var}(\bar{Y}) = \frac{(e^{\sigma^2_{\tilde{y}}} - 1)\, e^{2\mu_y + \sigma^2_{\tilde{y}}}}{M}, \tag{A.21}$$

so the (exact) coefficient of variation is

$$\mathrm{CV}(\bar{Y}) = \frac{\sqrt{\mathrm{Var}(\bar{Y})}}{\mathbb{E}(\bar{Y})} = \sqrt{\frac{e^{\sigma^2_{\tilde{y}}} - 1}{M}}, \tag{A.22}$$

which is (A.10). To obtain (A.11), we approximate $\bar{Y}$ by a log-normal variable $\widetilde{Y} \sim LN(\mu_{\bar{Y}}, \sigma^2_{\bar{Y}})$ matching the first two moments (Fenton–Wilkinson; see Marlow, 1967).[1] Solving the moment-matching conditions gives

$$\sigma^2_{\bar{Y}} = \ln\left[\frac{e^{\sigma^2_{\tilde{y}}} - 1}{M} + 1\right], \qquad \mu_{\bar{Y}} = \mu_y + \tfrac{1}{2}\sigma^2_{\tilde{y}} - \tfrac{1}{2}\sigma^2_{\bar{Y}}. \tag{A.23}$$

[1] The sum of i.i.d. log-normal variables is not log-normal in general, but is well-approximated by one with matched moments. The exact moments (A.21) hold regardless of this approximation; the approximation is only used to express $\ln \bar{Y}$ as an explicitly normal variable in (A.11).

Substituting $\mu_y = \mu_{\tilde{y}} + \frac{\nu}{1-\nu}\ln(\nu/\omega)$ and isolating the wage,

$$\mu_{\bar{Y}} = \underbrace{\mu_{\tilde{y}} + \tfrac{\nu}{1-\nu}\ln\nu + \tfrac{1}{2}\sigma^2_{\tilde{y}} - \tfrac{1}{2}\sigma^2_{\bar{Y}}}_{\text{constant}} - \tfrac{\nu}{1-\nu}\ln\omega, \tag{A.24}$$

which gives the representation (A.11) with $\zeta := \ln\bar{Y} - \mu_{\bar{Y}} \sim N(0, \sigma^2_{\bar{Y}})$. ■

B Frictionless Economy

In this Appendix, we show that with $\psi = 0$ the distribution μ is characterized by the distribution of firms across productivity transitions h and one capital level per previous productivity state.

In equation (1), the choice of the current level of employment can be derived from a static problem as:

$$N(\varepsilon, k; \mu) = \arg\max_n [\varepsilon k^\alpha n^\nu - \omega(\mu) n] \tag{B.1}$$

which yields

$$N(\varepsilon, k; \mu) = [\nu\varepsilon k^\alpha/\omega(\mu)]^{1/(1-\nu)} \tag{B.2}$$

Using this decision rule for employment, we can replace the first and second terms in equation (1) as:

$$\varepsilon k^\alpha n^\nu - \omega(\mu) n = (1-\nu)\varepsilon^{1/(1-\nu)} k^{\alpha/(1-\nu)} \left(\frac{\nu}{\omega(\mu)}\right)^{\nu/(1-\nu)}, \tag{B.3}$$

and we can rewrite the problem as follows:

$$\begin{aligned} v(\varepsilon, k; \mu) = \max_{k'} \Big[&(1-\nu)\varepsilon^{1/(1-\nu)} k^{\alpha/(1-\nu)} \left(\frac{\nu}{\omega(\mu)}\right)^{\nu/(1-\nu)} + (1-\delta)k - k' \\ &+ E\big[d(\mu, \mu') v(\varepsilon', k'; \mu') \mid \varepsilon, \mu\big]\Big]. \end{aligned} \tag{B.4}$$

This problem yields the optimal investment decision $G(\varepsilon; \mu)$ as follows:

$$G(\varepsilon; \mu) = \left(\frac{\alpha \sum_{\mu'} \Pi^\mu(\mu'|\mu)\, d(\mu, \mu') \left(\frac{\nu}{\omega(\mu')}\right)^{\nu/(1-\nu)} \sum_{\varepsilon'} \Pi^\varepsilon_{\mu'|\mu}(\varepsilon'|\varepsilon)\, \varepsilon'^{\,1/(1-\nu)}}{1 - (1-\delta)\sum_{\mu'} \Pi^\mu(\mu'|\mu)\, d(\mu, \mu')} \right)^{\frac{1-\nu}{1-\alpha-\nu}}. \tag{B.5}$$

This shows that the investment decision is independent of the current capital stock k, which depends only on the idiosyncratic productivity of the previous period and the aggregate state. This implies that (1) it is sufficient to track the idiosyncratic productivity both in the current and previous periods for each firm and (2) the distribution of the current and previous idiosyncratic productivity h is a N_ε by N_ε grid point object.

It follows that the distribution of firms over idiosyncratic productivity and capital stock can be recovered, $\mu(\varepsilon_j, k_i)$, from h and the capital levels $G(\varepsilon_i; \mu_{-1})$ in each period as follows. First, we can construct $\mu_{\varepsilon,-1}(\varepsilon_i)$, the marginal distribution of firms over ε_i for $i = 1, \ldots, N_\varepsilon$ in the previous period -1, and $\mu_\varepsilon(\varepsilon_j)$, the marginal distribution of firms over ε_j for $j = 1, \ldots, N_\varepsilon$ in the current period. We can also construct $\Pi_h^\varepsilon(\varepsilon_i, \varepsilon_j)$, the transition probability $Pr(\varepsilon = \varepsilon_j \mid \varepsilon_{-1} = \varepsilon_i)$. Therefore, we can construct $\mu(\varepsilon, k)$, the distribution of firms over productivity and stock of capital in each period as

$$\mu(\varepsilon_j, k_i) = \mu_{\varepsilon,-1}(\varepsilon_i)\Pi_h^\varepsilon(\varepsilon_i, \varepsilon_j) \tag{B.6}$$

$$k_i = G(\varepsilon_i; \mu_{-1}), \qquad K = \sum_i \mu_{\varepsilon,-1}(\varepsilon_i)\, G(\varepsilon_i; \mu_{-1}) \tag{B.7}$$

C Model Solution

In this section, we first outline the methodology used to construct the set of firm distributions across transitions and the corresponding probability space. We then describe the approach employed to solve the heterogeneous firm model.

C.1 Firm Distribution Across Transitions and its Probability Space

We construct the set of firm distributions across transitions, denoted by H, along with the associated transition probability $\Pi(h' \mid h)$, to replicate the fluctuations in the marginal distribution of firms over productivity, $\mu_\varepsilon(\varepsilon)$. To construct the transition matrix $\Pi(h' \mid h)$, we approximate the multivariate stochastic process governing the set of moments m^ε

of the distribution $\mu_\varepsilon(\varepsilon)$ using a finite-state Markov chain. We then use this discretized process, together with the ergodic idiosyncratic transition probability Π^ε, to construct the firm distribution across transitions, H.

C.1.1 Moments Selection and Construction of the transition probability $\Pi(h'|h)$

To make the problem computationally feasible, we construct the set of firm distributions across transitions, denoted by H, along with the associated transition probability $\Pi(h' \mid h)$, to replicate the fluctuations in the most volatile moments of the marginal productivity distribution, $\mu_\varepsilon(\varepsilon)$. To this end, after simulating $\mu_\varepsilon(\varepsilon)$, we implement the following steps:

1. Identify the granular segment of the productivity distribution—specifically, the region of $\mu_\varepsilon(\varepsilon)$ where the Law of Large Numbers fails.

2. Select the set of most volatile moments that allow us to replicate the fluctuations in the firm distribution within this granular segment.

3. Approximate the stochastic process governing the selected moments m^ε of $\mu_\varepsilon(\varepsilon)$ using a multivariate first-order Markov chain, following the discretization method of Terry and Knotek II (2011). This yields a transition matrix $\Pi^{m^\varepsilon}(m^{\varepsilon\prime} \mid m^\varepsilon)$.

The moment process m^ε is approximated by a first-order Markov chain, so the transition between two distributions across transitions depends only on their common marginal: for h with marginals $(\mu_{\varepsilon,p}, \mu_{\varepsilon,q})$ and h' with marginals $(\mu_{\varepsilon,q}, \mu_{\varepsilon,r})$, $\Pi^h(h' \mid h) = \Pi^{m^\varepsilon}(m^\varepsilon_r \mid m^\varepsilon_q)$, and $\Pi^h(h' \mid h) = 0$ for any other h'.

Figure C.1 presents the simulated productivity distributions, $\mu_\varepsilon(\varepsilon)$, on a logarithmic scale over 10,000 periods. The simulation indicates that the Law of Large Numbers does not hold in the right tail of the productivity distribution. Consequently, we define the granular section of the productivity distribution as the portion of $\mu_\varepsilon(\varepsilon)$ associated with $\varepsilon \geq \varepsilon_{13}$.

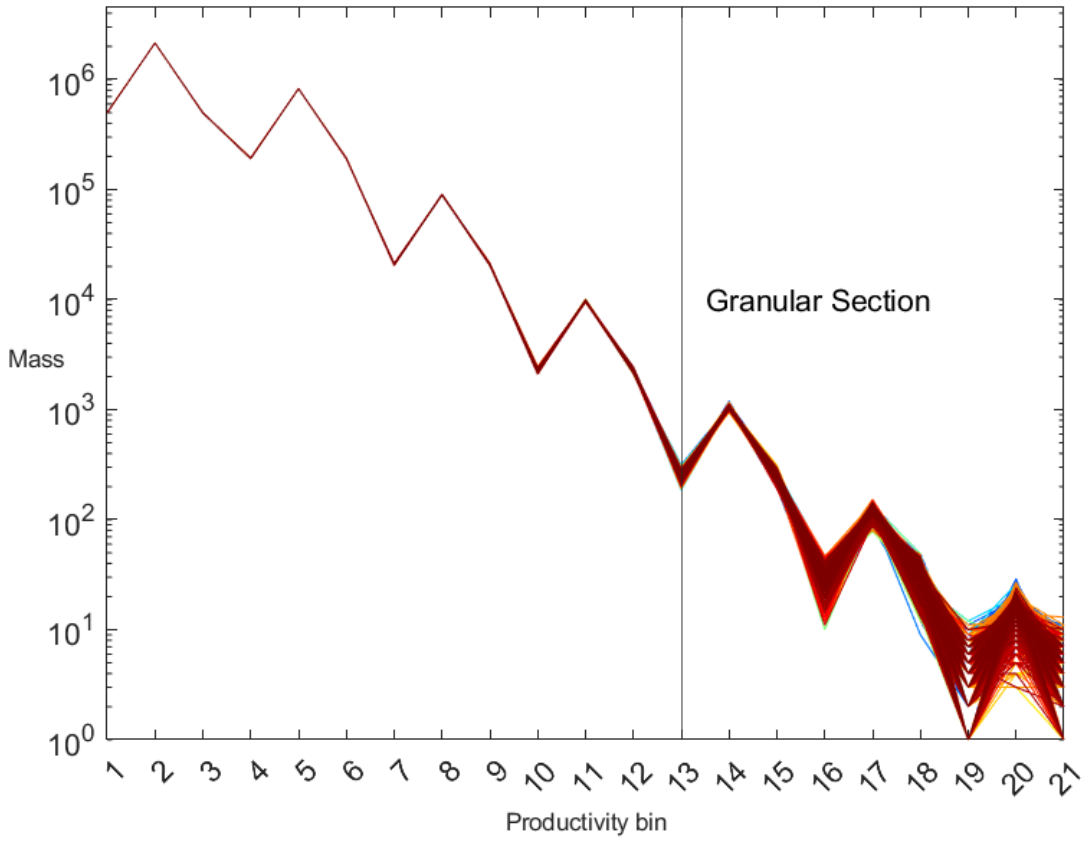
Next, Figure C.2 reports the distributions of the percentage deviations from their ergodic values for the moments that characterize fluctuations in the productivity distribution $\mu_\varepsilon(\varepsilon)$

within the granular section. Specifically, Figure C.2 displays the percentage deviations of: (A) mean productivity, (B) the standard deviation of productivity, (C) Pearson's moment coefficient of skewness, and (D) firm mass—each calculated among firms with $\varepsilon \geq \varepsilon_{13}$.

Interestingly, the simulation reveals that the dispersion of the percentage deviation in the mean (0.398) is relatively low compared to those of the standard deviation (5.655), skewness (8.632), and firm mass (2.999). Based on this, we simulate a granular economy that replicates the cyclicality of: (1) the standard deviation, (2) Pearson's moment coefficient of skewness, and (3) firm mass within the granular section of the productivity distribution.

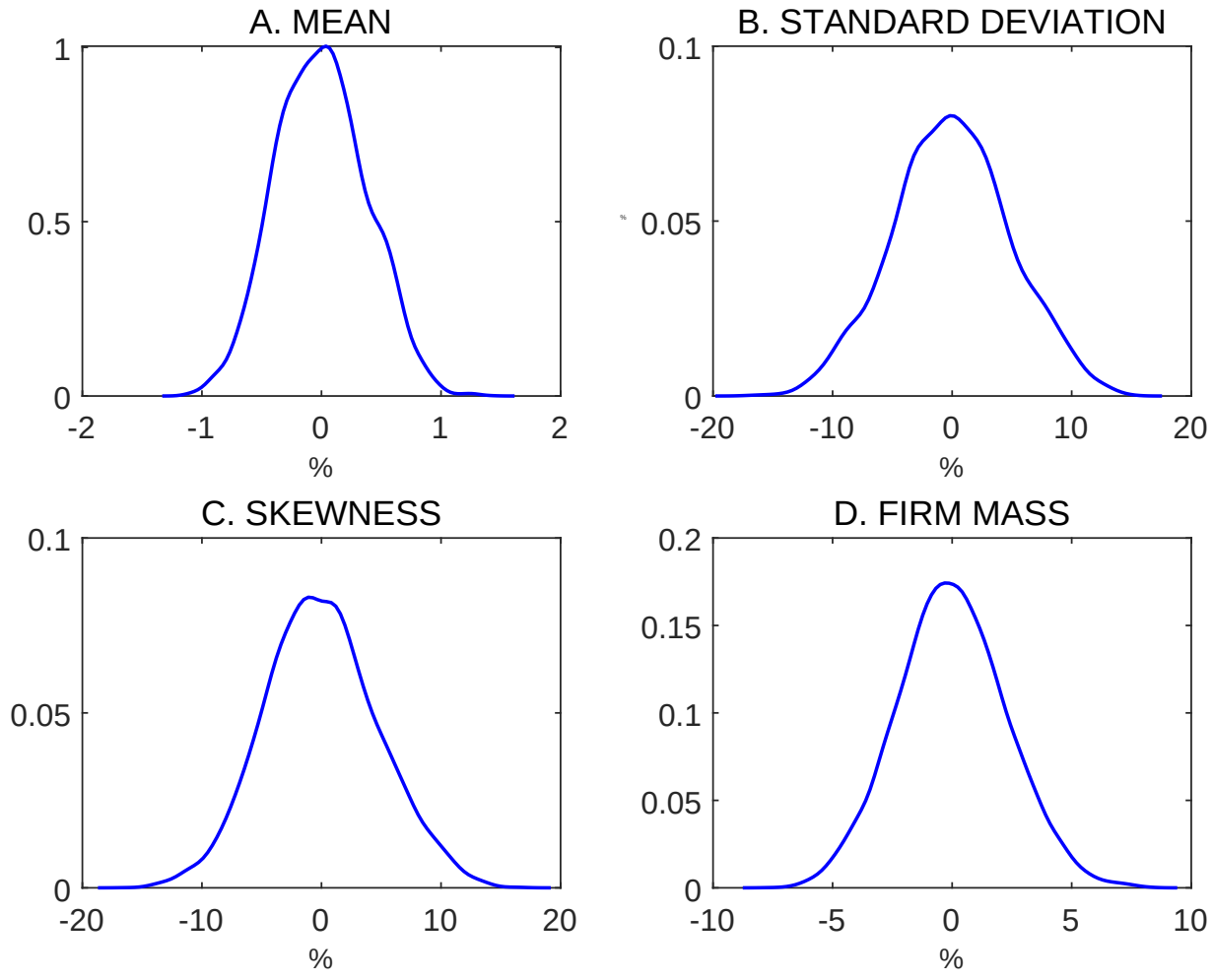
To discretize the multivariate stochastic process of the moment m^{ε}, we use a grid with three points for each moment, i.e., $d_{m^{\varepsilon}} = 3$.

Figure C.1: Simulated Productivity Distribution $\mu_{\varepsilon}(\varepsilon)$



Notes: This figure illustrates the simulated productivity distribution, $\mu_{\varepsilon}(\varepsilon)$, on a logarithmic scale over 10,000 periods. Specifically, it depicts the mass of firms corresponding to each productivity level.

Figure C.2: Moments of the Tail



Notes: The figure reports the distributions of percentage deviations from their ergodic values for: (A) mean productivity, (B) standard deviation of productivity, (C) Pearson's moment coefficient of skewness, and (D) firm mass—each calculated among firms with $\varepsilon \geq \varepsilon_{13}$. These distributions are derived from a 10,000-period simulation of the productivity distribution, $\mu_{\varepsilon}(\varepsilon)$.

C.1.2 Constructing H

We construct the set of firm distributions across transitions,

$$H = \{h_1, h_2, \ldots, h_{N_h-1}, h_{N_h}\},$$

by iterating steps 1 and 2 for F times, in order to ultimately implement step 3:

1. Construct

$$H^f = \{\mu^f_{\varepsilon,1}, \mu^f_{\varepsilon,2}, \ldots, \mu^f_{\varepsilon,N_h}\}$$

to reflect the discretized distribution m^{ε}.

2. Given Π^{ε}, implement a constrained draw of h such that each element of the set

$$H^f = \{h_1^f, h_2^f, \ldots, h_{l+(m^{\varepsilon}-1)N_{\mu_\varepsilon}}^f, h_{l+(m^{\varepsilon}-1)N_{\mu_\varepsilon}+1}^f, \ldots, h_{N_{\mu_\varepsilon}^2}^f\}$$

satisfies the following conditions for all $p, q = 1, 2, \ldots, N_{\mu_\varepsilon}$ and $i, j = 1, 2, \ldots, N_{\varepsilon}$:

$$\mu_{\varepsilon,q}^f(\varepsilon_i) = \sum_j h_{p+(q-1)N_{\mu_\varepsilon}}^f(\varepsilon_i, \varepsilon_j),$$
$$\mu_{\varepsilon,p}^f(\varepsilon_j) = \sum_i h_{p+(q-1)N_{\mu_\varepsilon}}^f(\varepsilon_i, \varepsilon_j),$$

where $N_{\mu_\varepsilon} = (d_{m^{\varepsilon}})^3$.

3. Compute the average distribution:

$$H = \frac{1}{F}\sum_{f=1}^{F} H^f.$$

C.2 Model Simulation

C.2.1 Firm's Problem

We can rewrite the firm's problem (Equation 1) using marginal utility, $p \equiv U_c'(c, n^h)$, as follows:

$$\widehat{v}(\varepsilon, k; \mu) = \max_{n,i} p(\mu)\left[\varepsilon F(k, n) - \omega(\mu)n - i\right] + \beta \sum_{\mu'} \Pi^{\mu}(\mu'|\mu) \sum_{\varepsilon'} \Pi_{\mu'|\mu}^{\varepsilon}(\varepsilon'|\varepsilon)\widehat{v}(\varepsilon', k'; \mu') \quad \text{(C.1)}$$

The evolution of the aggregate equilibrium is characterized by the following mappings:

$$\begin{aligned} p &= \Gamma_p(\mu), \\ \mu' &\sim G(\mu). \end{aligned} \quad \text{(C.2)}$$

Given the transition probabilities defined above, we express the Bellman equation in terms of h and K, explicitly substituting these for μ:

$$\begin{aligned} v(\varepsilon, k; h, K) = \max_{n,i} \; & p(h, K) \left[\varepsilon F(k, n) - \omega(h, K)n - i\right] \\ & + \beta \sum_{h'} \Pi^h(h'|h) \sum_{\varepsilon'} \Pi^{\varepsilon}_{h'}(\varepsilon'|\varepsilon) v(\varepsilon', k'; h', K') \end{aligned} \tag{C.3}$$

We approximate the equilibrium mappings using log-linear rules:

$$\ln(\widehat{p}) = \alpha_p(h) + \beta_p(h) \ln(K), \tag{C.4}$$

$$\ln(\widehat{K}') = \alpha_K(h) + \beta_K(h) \ln(K) \tag{C.5}$$

These log-linear rules are used to forecast the future values of the price level p and aggregate capital stock K'. The coefficients α and β depend on the current aggregate state h.

C.2.2 Solution Algorithm

The solution algorithm consists of an inner and outer loop:

- **Inner Loop:** Solve the firm's value function, $v(\varepsilon, k; h, K)$, via value function iteration, holding the forecasting rules for aggregate variables (Equations C.4) fixed.
- **Outer Loop:** Simulate the economy forward. In each period, draw a new aggregate state h' based on Π^h. Use the firm's policy functions (from the inner loop) and the transition probabilities $\Pi^{\varepsilon}_{h'}$ to determine firms' optimal decisions and update the aggregate capital stock and the distribution of firms. Then, update the forecasting rules.

The algorithm iterates between the inner and outer loops until the coefficients of the forecasting rules converge. This provides an approximate solution to the recursive equilibrium.

C.3 Performance

Our procedure to build firm distributions across transitions delivers an error of 0.0000830440, computed as follows:

$$\frac{1}{N_h}\sum_{j=1}^{N_h}\frac{1}{3}\left(\frac{\left|m_{1,j}^{\text{target}}-\text{st.dev.}(h_j)\right|}{m_{1,j}^{\text{target}}}+\frac{\left|m_{2,j}^{\text{target}}-\text{skew}(h_j)\right|}{m_{2,j}^{\text{target}}}+\frac{\left|m_{3,j}^{\text{target}}-\text{Tail Mass}(h_j)\right|}{m_{3,j}^{\text{target}}}\right) \tag{C.6}$$

Figure C.3 presents the kernel density estimates of the R^2 values for the 138 forecasting rules applied to price and capital. For the capital forecasting rules, the mean and minimum R^2 values are 0.9992 and 0.9891, respectively. Similarly, for the price forecasting rules, the mean and minimum R^2 values are 0.9992 and 0.9962.

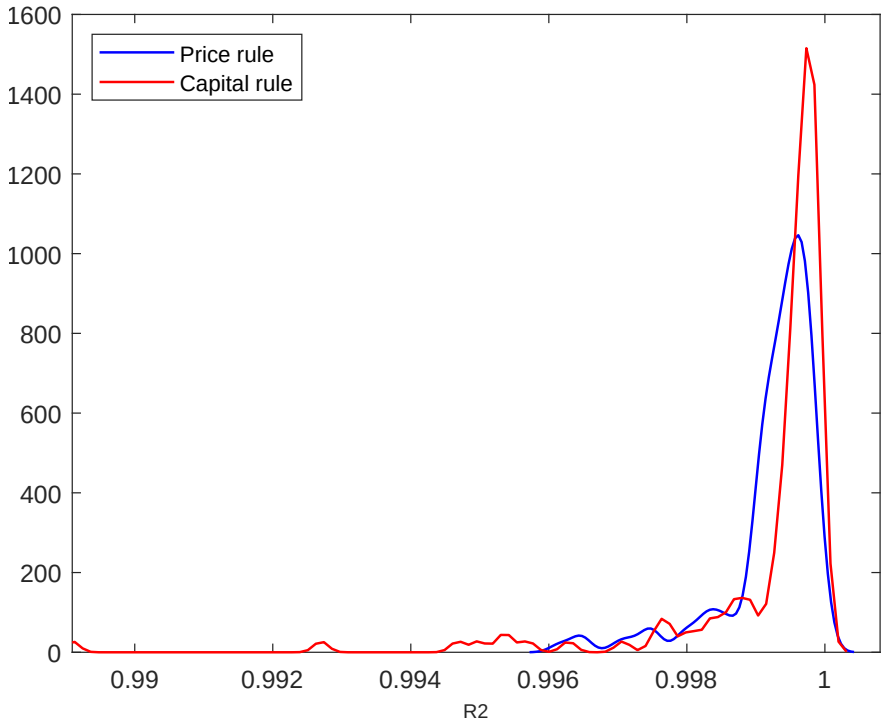


Figure C.3: Distribution of the R^2 of the Forecasting Rules

Notes: The figure reports the kernel densities of the R^2 values for the 138 forecasting rules. The blue line corresponds to price forecasting rules, and the red line to capital forecasting rules.

D Data

Sample Selection.— We construct an annual panel dataset by merging firm-level TFP estimates from İmrohoroğlu and Tüzel (2014) with Compustat data from 1964 to 2019. Using Standard Industry Classification (SIC) codes, we exclude firms in the oil, energy, and financial sectors from our sample. Specifically, we exclude:

- **Oil and oil-related firms:** SIC codes 2911, 5172, 1311, 4922, 4923, 4924, and 1389;
- **Energy firms:** SIC codes ranging from 4900 to 4940;
- **Financial firms:** SIC codes ranging from 6000 to 6999.

We also eliminate firms with missing data to ensure valid sales observations across the sample. The TFP estimates are adjusted using industry-specific time dummies to remove industry-year effects.

Variable Construction.—We define the variables used in our empirical analysis as follows:

1. *Gross investment rate:* Ratio of capital expenditures (CAPX) to plant, property, and equipment (PPEGT). We control for 2-digit sector-by-year fixed effects.
2. *Inaction rate:* Mass of firms such that $|i/k| \leq 0.01$.
3. *Net investment rate:* Growth rate of PPEGT. We control for 2-digit sector-by-year fixed effects.
4. *Negative investment spike:* Among the 500 largest firms (by sales) in the previous period, defined as the mass of firms such that
$$\Pr\left(\frac{\Delta k'}{k} \leq -0.2\right).$$
5. *Marginal product of capital* (MPK): Logarithm of the ratio of sales (SALE) to physical capital (PPEGT). We control for 2-digit sector-by-year fixed effects.

6. *Idiosyncratic shocks:* Approximated by the productivity growth rate:

$$\Delta\varepsilon_{i,t} = \frac{\varepsilon_{i,t} - \varepsilon_{i,t-1}}{\varepsilon_{i,t-1}}. \tag{D.1}$$

7. *Granular residual:* Difference between the sales-weighted and equally weighted average of idiosyncratic shocks among the 200 largest firms in the previous period:

$$\Theta_t^{\text{Gabaix}} = \sum_{i=1}^{100} \frac{\text{Sale}_{i,t-1}}{\text{Sale}_{t-1}^{100}} \Delta\varepsilon_{i,t} - \frac{1}{100} \sum_{i=1}^{100} \Delta\varepsilon_{i,t}. \tag{D.2}$$

8. *Granular capital misallocation:* Dispersion of MPK among the 200 largest firms:

$$\sigma_t^{\text{MPK},200} = \sqrt{\frac{1}{200} \sum_{i=1}^{200} (MPK_{i,t} - MPK_t^{\text{mean}})^2}. \tag{D.3}$$

Table D.1: Summary Statistics

x	Observations	mean	sd	min	max
Θ_t^{Gabaix}	38	0.000	0.017	-0.031	0.056
$\ln(\sigma^{MPK,200})$	39	0.000	0.026	-0.055	0.0771
$\ln(\sigma^{MPK,-200})$	39	0.000	0.017	-0.048	0.047
$Pr(\lvert\frac{i}{k}\rvert \leq 0.01)$ TOP 500	39	0.111	0.021	0.075	0.186
$Pr(\frac{\Delta k'}{k} \leq -0.2)$ TOP 500	39	0.024	0.0282	0.000	0.197

Notes: This table reports descriptive statistics for the variables used in Section 2. All series are HP-filtered with a smoothing parameter of 6.25, except for the inaction rate and the negative investment spike.

E Cyclicality of the Real Interest Rate over Time

Table E.1: Cyclical Dynamics of Risk-Free Rate

	$\sigma(\ln(1+r))$	$\rho(\ln(1+r), \ln Y)$
Whole sample	1.310	-0.286
Pre-1980	1.131	-0.828
Post-1980	1.178	0.189

Notes: Real interest rate is measured as the nominal return on 1-year Treasury bills (DGS1 taken from FRED) adjusted for realized CPI inflation (CPIAUCSL_PC1 taken from FRED). Output measured as real gross domestic product (GDPCA taken from FRED). All series are HP-filtered in logs with a smoothing parameter of 6.25, following Ravn and Uhlig (2002). Whole sample refers to the 1964–2019 series. Pre-1980 refers to the 1964:1979 sample. Post-1980 refers to the 1980–2019 sample.

F Decomposition of Irreversibility's Impact

Section 3.2 identifies two channels through which capital irreversibility dampens aggregate volatility in our granular economy. The direct allocation channel (Section 3.2.1) reduces the cross-sectional covariance between capital and productivity, lowering the cyclical alignment of capital across firms. The indirect general equilibrium channel (Section 3.2.3) operates through the consumption–wage link $\omega(\mu) = \eta C(\mu)$: by impeding investment, irreversibility raises consumption volatility, and the resulting wage flexibility cushions firm-level labor demand. This appendix quantifies the relative contributions of these two channels using two counterfactual economies.

The first counterfactual, *No Resource Loss*, retains the friction ψ in firms' optimization but sets the aggregate resource loss to zero, $\Psi \equiv \psi \int \hat{i}^{-}\, dm = 0$. Firms face the same friction as in the benchmark, so the direct allocation channel and the cross-sectional dispersion of marginal products of capital are unchanged. What is shut down is the cyclical drag of Ψ on the aggregate resource constraint: with no real resources destroyed at disinvestment, the household's consumption-smoothing capacity improves, $\sigma(\ln C)$ falls, and through $\omega = \eta C$ the wage becomes less responsive to shocks. The indirect general equilibrium channel is thus weakened relative to the benchmark.[2] This economy coincides with economy *(2-a: No resource loss)* of Section 4.

The second counterfactual, *Household Rents Capital*, eliminates factor misallocation by construction: the representative household rents capital to firms each period, so that the marginal product of capital is equalized across firms. The direct allocation channel is shut down by design; what remains of irreversibility's effect operates through the indirect general equilibrium channel and the resource-loss component of the household's budget constraint.

[2]This channel is attenuated, not switched off. Because the friction ψ continues to bind in firms' optimization, aggregate investment remains less responsive to shocks than under $\psi = 0$, and this residual constraint still passes through to consumption and wages.

Table F.1: Counterfactual Decomposition of Irreversibility's Volatility Effect

	Benchmark		*No Resource Loss*		*Household Rents Capital*	
	$\frac{\sigma(x)}{\sigma(x^{\psi=0})}$	$\rho(x, \ln Y)$	$\frac{\sigma(x)}{\sigma(x^{\psi=0})}$	$\rho(x, \ln Y)$	$\frac{\sigma(x)}{\sigma(x^{\psi=0})}$	$\rho(x, \ln Y)$
$\ln Y$	0.780	1.000	0.950	1.000	0.499	1.000
$\ln C$	1.429	0.785	1.050	0.702	0.882	0.702
$\ln I$	0.616	0.903	0.484	-0.098	0.386	0.967
$\ln H$	0.634	0.836	0.941	0.950	0.383	0.950
$\ln(1+r)$	1.390	-0.493	1.056	-0.294	0.889	0.823

Notes: The table reports business cycle moments for three economies: the *Benchmark*; the *No Resource Loss* counterfactual — equivalent to economy *(2-a)* of Section 4, in which firms face the friction ψ in their optimization but the aggregate resource loss $\Psi = \psi \int \hat{i}^{-} \, dm$ is set to zero; and the *Household Rents Capital* counterfactual, in which the representative household rents capital to firms each period, eliminating cross-firm misallocation. $\sigma(x)$ denotes the percentage standard deviation of variable x; $\sigma(x)/\sigma(x^{\psi=0})$ is the relative standard deviation with respect to its counterpart in the same economy with $\psi = 0$; and $\rho(x, \ln Y)$ is the contemporaneous correlation of x with output Y. All series are HP-filtered with a smoothing parameter of 6.25, following Ravn and Uhlig (2002).

Each column compares an economy under $\psi = 0.35$ to its own $\psi = 0$ counterpart, isolating the share of stabilization attributable to irreversibility within that environment. In the *Benchmark*, output volatility under $\psi = 0.35$ is 0.780 of the frictionless level—a 22 percentage-point reduction. In the *No Resource Loss* economy this stabilization collapses to roughly 5 percentage points (0.950), implying that the indirect general equilibrium channel and the resource-loss component together account for at least three-quarters of irreversibility's total dampening of output volatility. The corresponding muting of consumption volatility (from a ratio of 1.429 to 1.050) and of interest-rate volatility (from 1.390 to 1.056)

follows the same pattern, consistent with the consumption–wage link of Section 3.2.3.

In the *Household Rents Capital* economy, irreversibility cuts output volatility roughly in half (0.499) even though cross-firm misallocation is eliminated by construction. The direct allocation channel is therefore not the sole margin through which irreversibility stabilizes the granular economy: even with capital allocated efficiently across firms, the friction continues to constrain aggregate investment and to operate through the indirect general equilibrium channel.

G Welfare along the Transition

This appendix describes the computation of the second welfare row of Table 8.

The policy is imposed at date zero, unexpectedly and permanently. The economy starts from the Benchmark's ergodic distribution of firms over productivity and capital, $\mu(\varepsilon, k)$, and firms use the decision rules and the forecasting rules of the destination economy from the first period. Granular shocks are on throughout.

We simulate R windows of $T = 200$ periods, indexed by $r = 1, \ldots, R$. Window r takes its sequence of aggregate states from a contiguous segment of the Benchmark's long simulation starting at date $t_0(r)$, with the starting dates evenly spaced over the simulation. Its initial cross-section is the Benchmark's ergodic distribution, with the marginal distribution over productivity set to the one realized at $t_0(r)$. The Benchmark's own path over the same window gives the welfare of staying in the Benchmark under the same shocks.

Welfare in window r is

$$W_r = \sum_{t=1}^{T} \beta^{t-1} \left[\ln C_t + \eta \left(1 - N_t \right) \right], \tag{G.1}$$

and W_r^B is the same sum along the Benchmark path. The consumption-equivalent variation is

$$\mathrm{CEV}_r = \exp\left[(1-\beta) \left(W_r - W_r^B \right) \right] - 1, \tag{G.2}$$

and Table 8 reports its average across windows, $R = 1{,}000$ for *(2-b)* and *(3)* and $R = 150$ for *(2-a)*. The dispersion across windows is 0.28 percentage points for *(2-b)*, 0.22 for *(3)* and 0.27 for *(2-a)*, and the sign is the same in every window.

Figure G.1 shows the average paths of capital, consumption and hours across windows for the three transitions.

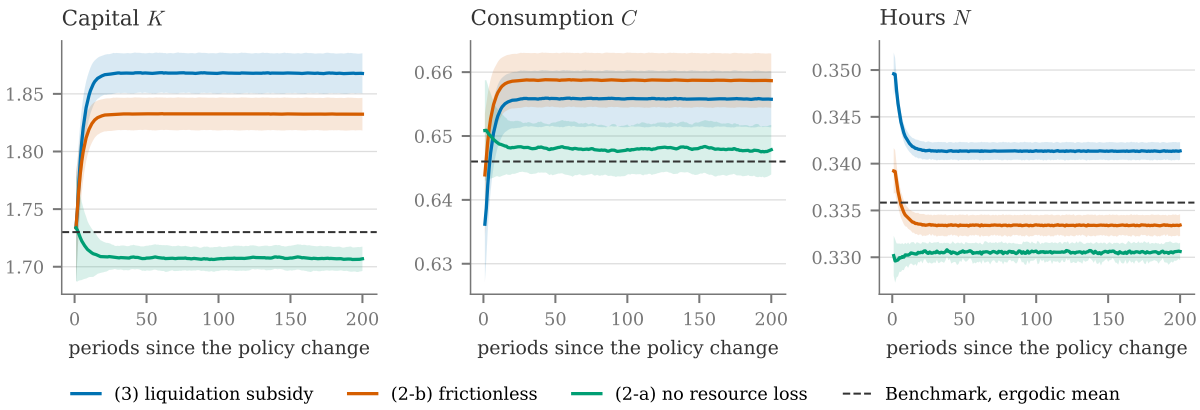


Figure G.1: Transition Paths after the Policy Change

Notes: Average across shock windows of the path after the policy change at date zero, starting from the Benchmark's ergodic distribution. Shaded areas are plus and minus one standard deviation across windows. The dashed line is the Benchmark's ergodic mean.

H Granular Shocks and Capital Misallocation in the data and model

Our theoretical framework argues for a significant role for granular shocks in driving capital misallocation, particularly when investment irreversibility is present. These dynamics, in turn, are argued to be important contributors to aggregate business cycle fluctuations. This section now presents empirical evidence from U.S. firm-level data to substantiate this narrative. We investigate the empirical linkage between identified granular shocks and ensuing changes in capital misallocation, and assess whether these observed dynamics align with the predictions and mechanisms of our structural model.

We first construct an annual panel dataset by merging the firm-level TFP estimates from İmrohoroğlu and Tüzel (2014) with Compustat databases from 1964 to 2019.[3] Similar to Gabaix (2011), we construct the series of granular residual as the sales-weighted idiosyncratic shocks among the 200 largest firms:[4]

$$\Theta_t^{Gabaix} = \sum_{i=1}^{100} \frac{\text{Sale}_{i,t-1}}{\text{Sale}_{t-1}^{100}} \Delta\varepsilon_{i,t} - \frac{\sum_{i=1}^{100} \Delta\varepsilon_{i,t}}{100} \tag{H.1}$$

where the proxy for idiosyncratic productivity shocks is the productivity growth rate $\Delta\varepsilon_{i,t} = \frac{\varepsilon_{i,t}-\varepsilon_{i,t-1}}{\varepsilon_{i,t-1}}$. To build a measure of capital misallocation, we follow Kilic and Tuzel (2025) and we approximate the marginal product of capital by the logarithm of the ratio of sales (SALE) to physical capital (PPEGT).[5] Accordingly, we measure capital misallocation using the dispersion of the marginal product of capital among the largest 200 firms, $\sigma^{MPK,200}$:

$$\sigma_t^{MPK,200} = \sqrt{\frac{\sum_{i=1}^{200}(MPK_{i,t} - MPK_t^{mean})^2}{200}} \tag{H.2}$$

Finally, we test whether the model can qualitatively and quantitatively replicate how gran-

[3] See Appendix D for the details of the variable constructions and Table D.1 therein for summary statistics.

[4] Note that we follow Gabaix and Koijen (2024) and we subtract the equal-weighted shocks from the weighted shocks to ensure the elimination of any common observed and unobserved aggregate factors.

[5] Idiosyncratic shocks and marginal product of capital are cleaned for 2-digit sector times year fixed effects.

ular misallocation responds to granular shocks. To this end, we employ Jordà (2005)'s local projection method to estimate impulse responses of granular misallocation to granular shocks. This method is based on the estimation of a series of regressions for each horizon τ for each variable. Accordingly, the linear model is as follows:

$$\ln(\sigma_{t+\tau}^{MPK,200}) = \alpha_\tau + \phi_\tau(L)z_{t-1} + \beta_\tau \Theta_t^{Gabaix} + v_{t+\tau} \quad \text{for} \quad \tau = 0, 1, 2, \ldots, \tag{H.3}$$

where z is a vector of control variables, $\phi_\tau(L)$ is a polynomial in the lag operator, and the shock is the identified granular shock. The shock is the granular residual. Our vector of baseline control variables, z, contains the logarithm of real GDP, 10-year T-bill, term spread (the government 5-year bond yield minus the 1-year yield), consumer price inflation, and the logarithm of the dispersion of the marginal product of capital computed excluding the largest 200.[6] Furthermore, z includes lags of the granular shock variable to control for any serial correlation in the shock variable. The term $\phi_\tau(L)$ is a first-order polynomial.

Figure H.1 compares the impulse responses of the logarithm of the misallocation of granular capital to a negative granular residual of one standard deviation between the model and the data. The model performs well in quantitatively replicating the response of capital misallocation among large firms to granular shocks, as the model IRFs lie within the 95 percent confidence interval of the empirical IRFs from the first year after the shock. A negative granular shock is associated with negative productivity shocks to larger firms, increasing capital misallocation. Importantly, the discrepancy between the IRFs generated by the model and those observed in the data is likely to arise from the model's exclusion of additional frictions that are present in the real economy. These omitted frictions contribute to a higher ergodic dispersion of the marginal product of capital, while simultaneously dampening its responsiveness to granular shocks. As shown in Figures H.2–H.4 below, this result is robust across

[6] In particular, we use the annual real GDP (GDPCA from FRED). We use yield treasury securities at 10, 5, and 1-year constant maturity (DGS10, DGS5 and DGS1 from FRED). We use consumer price inflation (FPCPITOTLZGUSA from FRED). The dispersion of marginal product of capital excluding the largest 100 firms is obtained from the COMPUSTAT dataset. All series are HP-filtered using a smoothing parameter of 6.25, following Ravn and Uhlig (2002).

alternative sample periods and industry-level controls used in the construction of firm-level MPK.

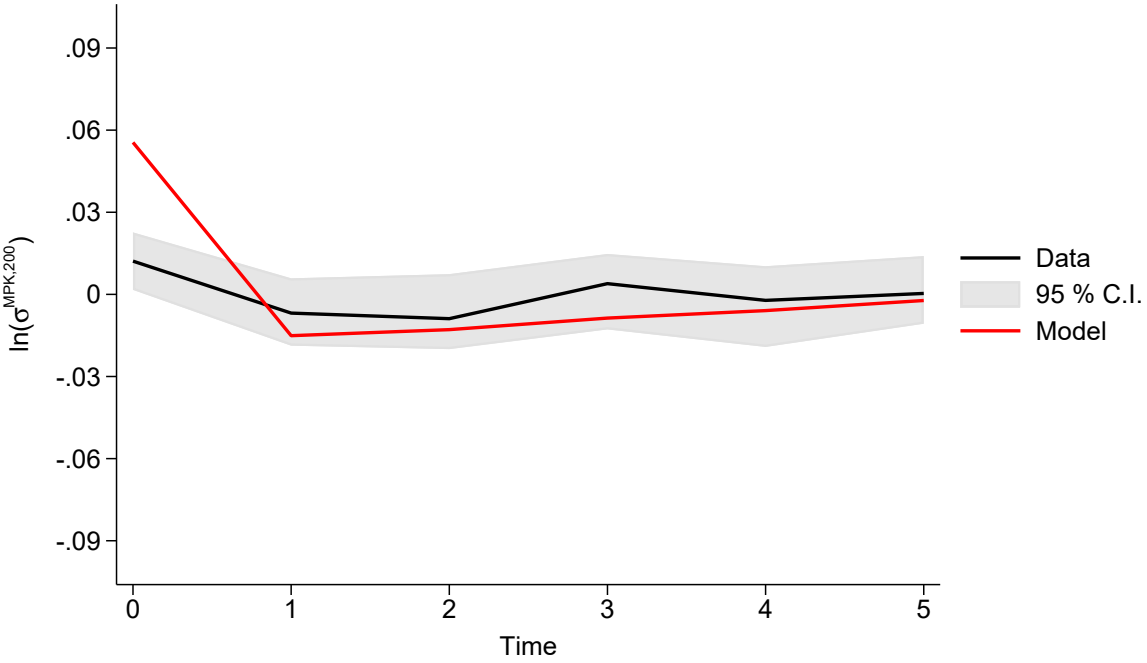
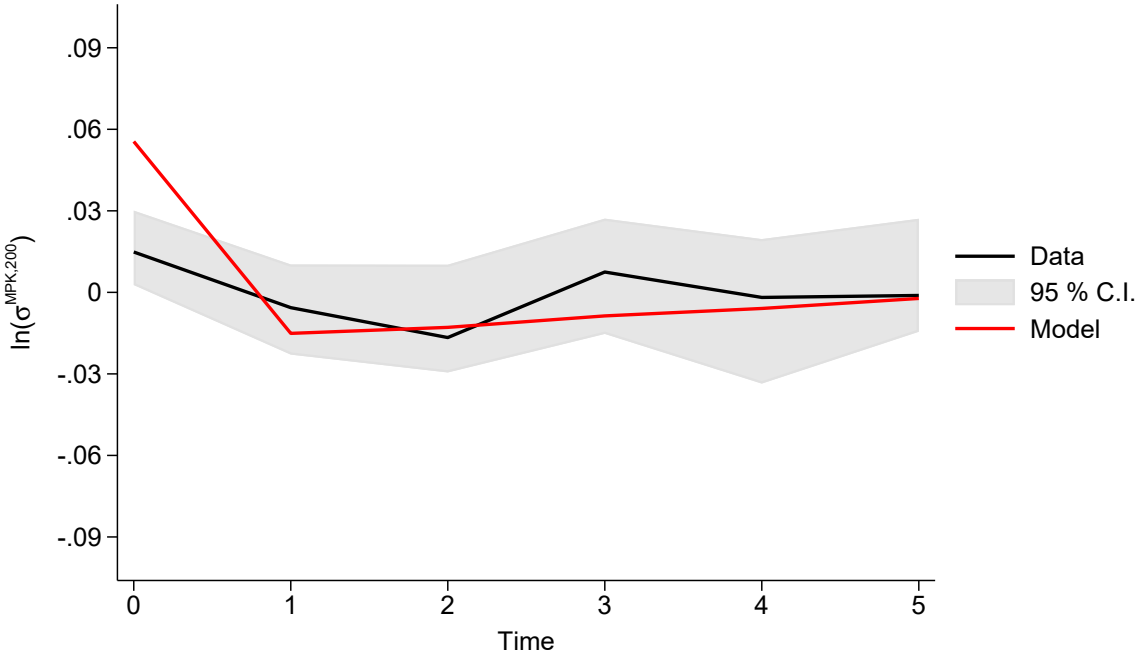


Figure H.1: Effect of granular residual shock on granular misallocation.

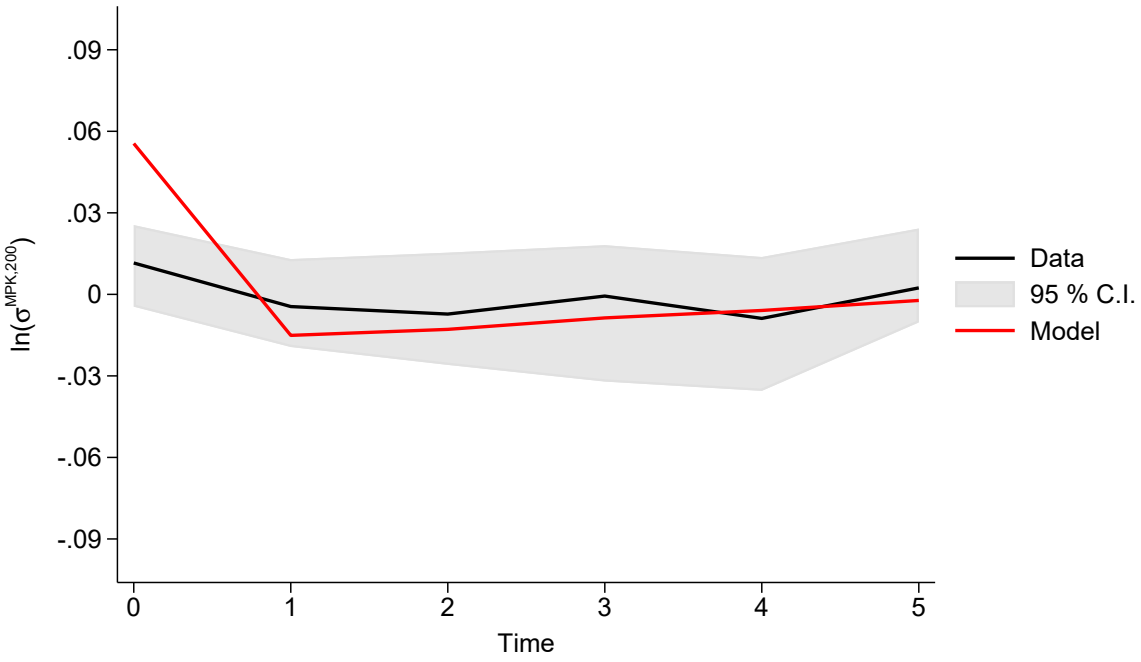
Notes: Responses of the logarithm of the granular dispersion of the marginal product of capital to a negative one standard deviation granular residual shock. The black line represents the response estimated from the data. The red line represents the response obtained from the simulated data. The 95 percent confidence intervals of the empirical estimations are computed with a wild bootstrap of 1000 repetitions. All series are HP-filtered using a smoothing parameter of 6.25, following Ravn and Uhlig (2002). Granular misallocation is estimated conditional on 2-digit sector-by-year fixed effects. For further details on data construction, see Appendix D.

Figure H.2: Effect of Granular Residual Shock on Granular Misallocation



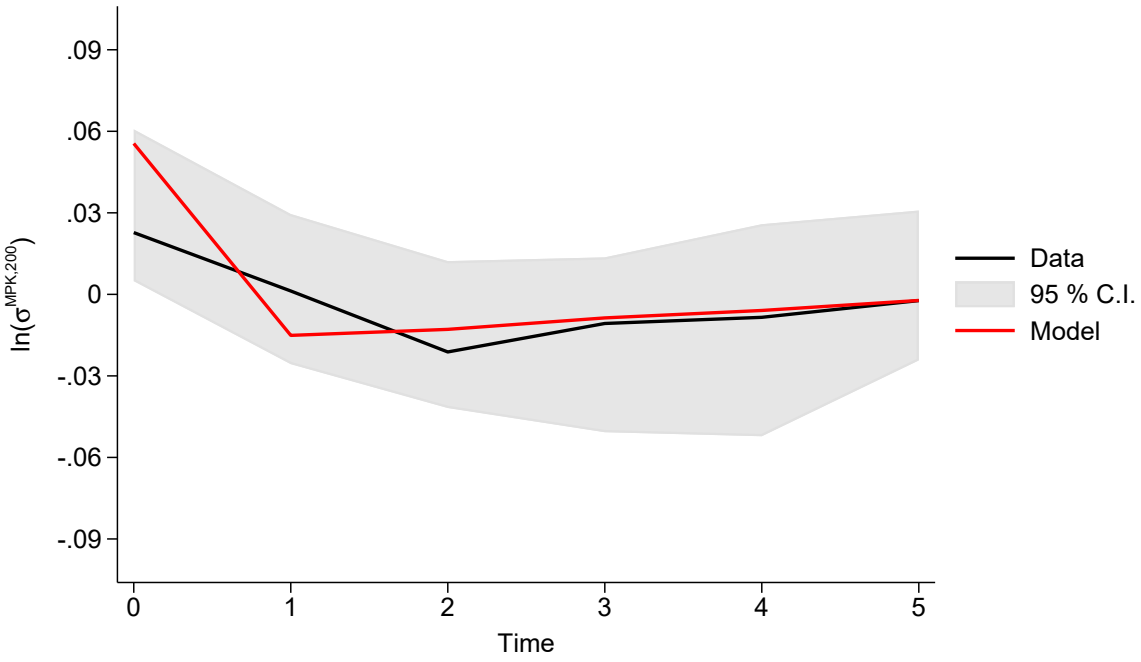
Notes: Responses of the logarithm of the granular dispersion of the marginal product of capital to a negative one standard deviation granular residual shock. The black line represents the response estimated from the data. The red line represents the response obtained from the simulated data. The 95 percent confidence intervals of the empirical estimations are computed with a wild bootstrap of 1000 repetitions. All series are HP-filtered using a smoothing parameter of 6.25, following Ravn and Uhlig (2002). The sample period spans from 1980 to 2019. Granular misallocation is estimated conditional on 2-digit sector-by-year fixed effects. For further details on data construction, see Appendix D.

Figure H.3: Effect of Granular Residual Shock on Granular Misallocation



Notes: Responses of the logarithm of the granular dispersion of the marginal product of capital to a negative one standard deviation granular residual shock. The black line represents the response estimated from the data. The red line represents the response obtained from the simulated data. The 95 percent confidence intervals of the empirical estimations are computed with a wild bootstrap of 1000 repetitions. All series are HP-filtered using a smoothing parameter of 6.25, following Ravn and Uhlig (2002). The sample period spans from 1964 to 2019. Granular misallocation is estimated conditional on 3-digit sector-by-year fixed effects. For further details on data construction, see Appendix D.

Figure H.4: Effect of Granular Residual Shock on Granular Misallocation



Notes: Responses of the logarithm of the granular dispersion of the marginal product of capital to a negative one standard deviation granular residual shock. The black line represents the response estimated from the data. The red line represents the response obtained from the simulated data. The 95 percent confidence intervals of the empirical estimations are computed with a wild bootstrap of 1000 repetitions. All series are HP-filtered using a smoothing parameter of 6.25, following Ravn and Uhlig (2002). The sample refers to the period spanning from 1980 to 2019. Granular misallocation is estimated conditional on 3-digit sector-by-year fixed effects. For further details on data construction, see Appendix D.

References

Gabaix, X., 2011. The Granular Origins of Aggregate Fluctuations. Econometrica 79, 733–772.

Gabaix, X., Koijen, R.S.J., 2024. Granular instrumental variables. Journal of Political Economy 132, 2274–2303.

İmrohoroğlu, A., Tüzel, Ş., 2014. Firm-level productivity, risk, and return. Management Science 60, 2073–2090.

Jordà, Ò., 2005. Estimation and inference of impulse responses by local projections. American Economic Review 95, 161–182.

Kilic, M., Tuzel, S., 2025. Investing in misallocation. Available at SSRN 4464542.

Ravn, M.O., Uhlig, H., 2002. On adjusting the Hodrick-Prescott filter for the frequency of observations. The Review of Economics and Statistics 84, 371–375.

Terry, S.J., Knotek II, E.S., 2011. Markov-chain approximations of vector autoregressions: Application of general multivariate-normal integration techniques. Economics Letters 110, 4–6.